

KARESUANDO, Sweden

WMO: 020810

Lat: 68.4419N

Lon: 22.4436E

Elev: 332

StdP: 97.40

Time Zone: 1.00 (EUC)

Period: 96-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-34.8	-31.6	-37.8	0.1	-34.4	-34.3	0.2	-31.4	11.8	-4.6	9.6	-4.0	1.0	320	0.752

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	8.8	24.0	15.3	21.7	14.3	19.4	13.3	16.7	21.1	15.4	19.5	14.3	18.2	3.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
15.0	11.1	17.7	13.7	10.2	16.9	12.5	9.4	15.8	48.0	21.2	44.3	19.3	41.1	18.2	21.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.7	7.5	6.3	DB	-38.0	26.3	4.3	1.9	-41.1	27.6	-43.6	28.7	-46.0	29.8	-49.1	31.2	(4) (5)
(5)			WB	-37.3	18.1	3.9	1.6	-40.1	19.2	-42.4	20.2	-44.6	21.1	-47.5	22.3	(4) (5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	-1.2	-13.8	-14.0	-9.0	-2.7	4.1	10.1	13.9	10.9	6.2	-1.0	-8.1	-11.6	(6)
(7)		DBStd	11.30	8.50	8.64	6.45	4.82	4.23	3.93	3.41	3.26	3.56	4.76	6.66	8.02	(7)
(8)		HDD10.0	4327	739	671	587	382	191	47	6	28	123	341	543	669	(8)
(9)		HDD18.3	7128	997	904	846	632	443	248	142	230	366	599	793	927	(9)
(10)		CDD10.0	247	0	0	0	0	7	49	127	57	7	0	0	0	(10)
(11)		CDD18.3	7	0	0	0	0	0	1	5	1	0	0	0	0	(11)
(12)		CDH23.3	79	0	0	0	0	5	14	53	7	0	0	0	0	(12)
(13)		CDH26.7	5	0	0	0	0	0	1	4	0	0	0	0	0	(13)
(14)	Wind	WSAvg	2.7	2.4	2.5	3.0	2.8	3.1	3.3	2.7	2.5	2.8	2.5	2.2	2.5	(14)
(15)	Precipitation	PrecAvg	458	25	22	23	24	32	49	75	63	43	42	34	26	(15)
(16)		PrecMax	666	48	65	61	71	81	114	214	152	95	104	81	71	(16)
(17)		PrecMin	268	4	2	2	3	3	9	12	12	7	8	4	4	(17)
(18)		PrecStd	87	10	14	12	14	19	24	38	31	24	22	16	14	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	3.7	3.7	5.0	10.3	22.9	25.2	27.0	24.4	17.3	10.0	5.0	4.3	(19)
(20)			MCWB	2.0	1.6	2.2	5.3	12.9	15.0	16.9	15.9	11.4	8.3	3.9	2.3	(20)
(21)		2%	DB	0.8	1.4	2.9	7.5	17.5	22.2	24.8	21.6	15.1	8.6	2.8	2.2	(21)
(22)			MCWB	-0.5	-0.2	0.8	3.6	10.1	13.3	15.9	14.9	11.2	7.1	2.0	0.9	(22)
(23)		5%	DB	-1.1	-0.8	1.4	5.7	13.6	19.6	22.7	19.0	13.4	7.1	1.3	0.3	(23)
(24)			MCWB	-1.9	-1.8	-0.4	2.5	8.0	12.1	15.1	13.7	10.3	5.7	0.8	-0.5	(24)
(25)		10%	DB	-3.2	-2.7	-0.3	4.1	10.8	17.0	20.3	17.1	11.9	5.2	0.2	-1.2	(25)
(26)			MCWB	-3.8	-3.5	-1.7	1.6	6.4	11.3	14.2	13.0	9.5	4.1	-0.2	-1.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	1.9	2.0	2.3	6.0	13.2	16.6	18.6	17.9	13.4	8.9	4.0	2.5	(27)
(28)			MCDWB	3.7	3.6	4.6	9.8	21.4	21.9	24.0	22.2	15.4	9.5	5.0	4.0	(28)
(29)		2%	WB	-0.3	-0.2	1.0	4.0	10.6	14.6	17.2	16.0	11.9	7.3	2.0	1.0	(29)
(30)			MCDWB	0.6	1.2	2.7	6.9	16.5	19.2	22.1	19.4	14.0	8.4	2.6	2.1	(30)
(31)		5%	WB	-2.0	-1.8	-0.3	2.8	8.5	13.3	16.1	14.6	10.9	5.8	0.8	-0.4	(31)
(32)			MCDWB	-1.1	-0.8	1.2	5.2	12.9	18.2	20.5	17.8	12.8	7.0	1.3	0.3	(32)
(33)		10%	WB	-3.8	-3.4	-1.6	1.8	6.9	11.9	15.0	13.4	9.9	4.3	-0.1	-1.8	(33)
(34)			MCDWB	-3.3	-2.7	-0.3	3.9	10.4	16.2	19.1	16.6	11.7	5.3	0.2	-1.1	(34)
(35)	Mean Daily Temperature Range	MDBR	8.6	8.9	10.0	9.3	8.1	8.4	8.8	8.8	7.6	5.8	7.1	8.2	(35)	
(36)		5% DB	MCDBR	8.4	7.9	8.1	9.5	13.2	12.6	12.5	12.6	10.0	6.4	5.0	6.6	(36)
(37)			MCWBR	7.8	7.0	6.5	6.2	7.0	5.7	5.0	6.5	6.2	4.9	4.7	6.1	(37)
(38)			MCDBR	8.0	7.6	7.9	8.6	12.0	10.8	10.6	10.7	8.0	5.8	4.9	6.6	(38)
(39)		5% WB	MCWBR	7.4	6.9	6.4	5.7	6.6	5.5	5.2	6.1	5.3	4.7	4.6	6.2	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.180	0.237	0.276	0.305	0.325	0.330	0.336	0.322	0.290	0.254	N/A	N/A	(40)	
(41)		taud	1.920	2.096	2.194	2.225	2.405	2.492	2.488	2.509	2.564	2.305	N/A	N/A	(41)	
(42)		Ebn at Noon	87	554	756	839	867	873	853	827	771	545	N/A	0	(42)	
(43)		Edh at Noon	7	45	76	100	95	90	88	77	55	36	N/A	0	(43)	
(44)	All-Sky Solar Radiation	RadAvg	0.03	0.47	1.97	3.97	5.04	4.93	4.63	3.29	1.77	0.58	0.08	0.00	(44)	
(45)		RadStd	0.00	0.04	0.09	0.18	0.49	0.35	0.45	0.23	0.18	0.09	0.01	0.00	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (38 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page